

Mingyu Kang

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PROFESSIONAL SUMMARY

Quantum physicist with experience spanning three major layers of the quantum computing stack: hardware, error correction, and applications. Research conducted at the theory-experiment interface – connecting atomic physics, quantum control, quantum error correction, and quantum chemistry – across academia and industry. First or co-first author on 8 of 12 peer-reviewed publications; developed open-source quantum software with >55 GitHub stars.

EDUCATION

Duke University <i>Ph.D. in Physics</i> • Dissertation: <i>Towards Quantum Advantage with Trapped Ions</i>	2020 – 2025 <i>Durham, NC</i>
Stanford University <i>B.S. Physics, with Distinction and Departmental Honors · GPA: 4.02/4.3</i>	2014 – 2020 <i>Stanford, CA</i>

EXPERIENCE

Postdoctoral Scholar <i>UC Berkeley, Dept. of Chemistry · Supervisor: Prof. K. Birgitta Whaley</i> • Performing resource estimation for early-fault-tolerant quantum chemistry algorithms, targeting practical quantum advantage benchmarks	Aug. 2025 – Present
Intern <i>QuEra Computing</i> • Modeled two-qubit gate error sources in neutral-atom processors using detailed atomic physics and system noise models, providing quantitative guidance for hardware improvement	Jun. 2024 – Sep. 2024
Intern (virtual) <i>IonQ Inc.</i> • Developed efficient protocols for high-precision motional-mode characterization in trapped-ion quantum computers, reducing calibration overhead (Quantum Science and Technology, 2023; patent pending)	Mar. 2021 – Sep. 2021
Ph.D. Researcher <i>Duke University, Dept. of Physics & Duke Quantum Center · Advisor: Prof. Kenneth R. Brown</i> • Designed pulse-optimization algorithms that improved two-qubit gate fidelities on trapped-ion processors; results experimentally validated (Physical Review Applied, 2021 & 2023; patent granted) • Developed erasure-conversion quantum error correction scheme exploiting trapped-ion error structure, achieving significantly higher fault-tolerance thresholds (PRX Quantum, 2023) • Co-designed and led trapped-ion analog quantum simulation experiments of molecular energy transfer dynamics (Nature Reviews Chemistry, 2024; Nature Communications, 2025)	Aug. 2020 – May 2025
Undergraduate Researcher <i>Stanford University · Advisor: Prof. Amir Safavi-Naeini</i> • Applied optimal control to maximize Bell-state fidelity in a nanomechanical quantum memory coupled to a superconducting qubit	Jan. 2019 – Jun. 2020
Sergeant & Signal Analyst <i>ROK Army SEC Security Lab</i> • Built GUI application for encoding, decoding, and identifying convolutional error-correcting codes of various types	Sep. 2016 – Jun. 2018

OPEN-SOURCE PROJECTS

QUITS: A modular Qldpc code circUIT Simulator github.com/mkangquantum/quits • Led development of a modular, flexible circuit-level simulator for quantum low-density parity check (LDPC) codes; resulted in a peer-reviewed publication (Quantum, 2025); earned >55 GitHub stars	2025
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AWARDS & GRANTS

- NSF Quantum Leap Challenge Institute Seed Grant** 2023 – 2024
Simulating Dissipative Electron Transfer with Trapped-Ion Systems
- \$100,000 research funding; collaboration with four other PhD students
- John T. Chambers Fellowship, Duke Fitzpatrick Institute for Photonics** 2020 – 2021
\$10,000; one of four awardees at Duke University
- Kwanjeong Scholarship, Kwanjeong Educational Foundation** 2014–2016, 2018–2020
\$55,000/yr; one of seven undergraduate awardees in South Korea

PUBLICATIONS

Published Journal Articles

- **MK***, Y. Zhang*, K. R. Brown, and T. Barthel, *Non-Gaussian Phase Transition and Cascade of Instabilities in the Dissipative Quantum Rabi Model*, Phys. Rev. A **113**, L061703 (2026). arXiv:2507.07092
- H. Ren, Y. Zhang, W. M. Billings, R. Tomann, N. V. Tkachenko, **MK**, M. Head-Gordon, and K. B. Whaley, *Error-mitigated nonorthogonal quantum eigensolver via shadow tomography*, Phys. Rev. Res. **8**, 013268 (2026). arXiv:2504.16008
- **MK***, Y. Lin*, H. Yao, M. Gökduman, A. Meinking, and K. R. Brown, *QUITS: A modular Qldpc code circUIT Simulator*, Quantum **9**, 1931 (2025). arXiv:2504.02673
- K. Sun*, **MK***, H. Nuomin, G. Schwartz, D. N. Beratan, K. R. Brown, and J. Kim, *Quantum Simulation of Spin-Boson Models with Structured Bath*, Nat. Commun. **16**, 4042 (2025). arXiv:2405.14624
- **MK**, H. Nuomin, S. N. Chowdhury, J. L. Yuly, K. Sun, J. Whitlow, J. Valdiviezo, Z. Zhang, P. Zhang, D. N. Beratan, and K. R. Brown, *Seeking a Quantum Advantage with Trapped-Ion Quantum Simulations of Condensed-Phase Chemical Dynamics*, Nat. Rev. Chem. **8**, 340–358 (2024). arXiv:2305.03156
- Q. Liang, **MK**, M. Li, and Y. Nam, *Pulse Optimization for High-Precision Motional-Mode Characterization in Trapped-Ion Quantum Computers*, Quantum Sci. Technol. **9**, 035007 (2024). arXiv:2307.15841
- **MK**, W. C. Campbell, and K. R. Brown, *Quantum Error Correction with Metastable States of Trapped Ions using Erasure Conversion*, PRX Quantum **4**, 020358 (2023). arXiv:2210.15024
- K. Sun, C. Fang, **MK**, Z. Zhang, P. Zhang, D. N. Beratan, K. R. Brown, and J. Kim, *Quantum Simulation of Polarized Light-Induced Electron Transfer with a Trapped-Ion Qutrit System*, J. Phys. Chem. Lett. **14**, 6071–6077 (2023). arXiv:2304.12247
- Z. Jia, S. Huang, **MK**, K. Sun, R. F. Spivey, J. Kim, and K. R. Brown, *Angle-Robust Two-Qubit Gates in a Linear Ion Crystal*, Phys. Rev. A **107**, 032617 (2023). arXiv:2210.04814
- **MK**, Q. Liang, M. Li, and Y. Nam, *Efficient Motional-Mode Characterization for High-Fidelity Trapped-Ion Quantum Computing*, Quantum Sci. Technol. **8**, 024002 (2023). arXiv:2206.04212
- **MK**, Y. Wang, C. Fang, B. Zhang, O. Khosravani, J. Kim, and K. R. Brown, *Designing Filter Functions of Frequency-Modulated Pulses for High-Fidelity Two-Qubit Gates in Ion Chains*, Phys. Rev. Applied **19**, 014014 (2023). arXiv:2206.10850
- **MK**, Q. Liang, B. Zhang, S. Huang, Y. Wang, C. Fang, J. Kim, and K. R. Brown, *Batch Optimization of Frequency-Modulated Pulses for Robust Two-Qubit Gates in Ion Chains*, Phys. Rev. Applied **16**, 024039 (2021). arXiv:2104.06887

SERVICE & OUTREACH

- Editorial Board, *Physical Review A*** Jan. 2026 – Dec. 2027
Early Career member
- Journal Referee**
Entropy, Nature Communications, npj Quantum Information, Physical Review A, Physical Review Letters, Physical Review Research, Physical Review X, PRX Quantum, Physica Scripta, Quantum, Science Advances
- Co-host, YouTube series *Quantum News Monthly*** Sep. 2023 – Sep. 2024
- Reviewed selected papers published each month in the field of quantum physics and computing